Magneto-optical trapping (MOT) uses fine-tuned lasers to cool and slow down atoms in a magnetic field that pushes the atoms to the center of the trap, essentially isolating the atoms and keeping them in a stable state which allows for atom interferometry–which uses the fact that atoms exhibit wave-like properties (they act as both waves and particles) to make very precise measurements. The Kovachy group uses this system to look for new short-range forces that exist beyond the Standard Model of physics. To fine-tune the lasers, the group relies on a method called saturated absorption spectroscopy (SAS), where light is passed through a cloud of gaseous Strontium (Sr) contained in a vapor cell (a heated, vacuum-sealed environment that holds gaseous atoms). On the other side of this cell is a detector that tells us how much light passes through. However, in the current system, Sr has begun to build up on the windows of the vapor cell interfering with SAS, so I will work with the group to build and test a new system to prevent this Sr build-up.

Atoms have what is called an excited and ground state, and each has a precise amount of energy associated with it. Photons from light can provide the energy required for a jump from the ground to excited state, but this requires the light to be at a very specific frequency to match the particular energy associated with this jump/transition. In SAS, if the laser matches the frequency needed, the atom will absorb the photons, and the detector will report little to no light passing through the vapor cell, and this will ensure that the laser is properly set to the frequency required for MOT. However, due to the higher temperatures required for Sr to remain in a gaseous phase, it is common for it to solidify near the ends of the vapor cell on the windows. So the new system I'll work to build and test will utilize sapphire heated windows and a long, baffled tube design to properly distribute heat throughout the vapor cell and prevent Sr build-up on the windows. Additionally, the other end goal of this project is to take what I learned from Kovachy's lab with SAS and apply it to a similar project using Rubidium (Rb) in Professor Odom's lab.

Many labs in the past have used Rb or Cesium (Cs) in vapor cells to lock lasers at the frequency they want. Rb, in particular, has been used in many applications, including atomic clocks, where the exact frequency of light is very important. In order to lock the laser at the frequency required, previous researchers have used SAS on Rb vapor cells.[1] Rb differs from Sr, however, because it can become a vapor at a much lower temperature, so it is easier to keep in a gaseous phase and doesn't easily build up on the windows. This has led to the use of Sr vapor cells being much less common.

However, some researchers have still used Sr. For example, Stellmer has provided a way to construct Sr vapor cells for frequency locking that significantly decreases the amount of Sr that sticks to the windows.[3] He describes that the use of a long vapor cell with a small diameter has been shown to successfully prevent Sr build-up on the windows for tests lasting over two years. However, while most systems, including Stellmer's, use a buffer gas (some inert gas that won't react with Sr) to push the Sr near the center of the cell and prevent Sr build-up, we don't have the tools necessary to easily fill our vapor cell with a buffer gas.

Therefore, our system will instead rely on a baffle design, which will help distribute heat over the cell, and heated sapphire windows to prevent Sr build up, both of which are new techniques in this lab that we hope to test and implement. Sapphire windows in vapor cells have been shown to be able to withstand higher temperatures without Rb or Sr sticking (500 ^{o}C for Rb and 580 ^{o}C for Sr, both more than high enough to keep Rb and Sr in gaseous phase), making it a good choice in our design.[2] The overall system should keep Sr heated and prevent it from cooling and sticking to the walls or windows of our vapor cell.

The early stage of my summer project will primarily consist of building and assembling the cell. After the strontium vapor cell is constructed, I will use SAS to make sure we are getting the absorption data we expect from the cell. To do this, we will set up a laser set at 461 nm (associated with the frequency we want for the transition state we want for Sr) to shine through the cell with a photodiode detector, which gives off an electrical signal when hit with light. We will then vary the frequency of the laser by small amounts, and the photodiode detector will tell us when light is or is not passing through the cell. We'll use this data to plot how far the laser frequency is from the resonant frequency (associated with 461 nm in our case) on the x-axis and plot how much light gets through the cell on the y-axis. We will compare this to a theoretical model where at x = 0 (laser is set at 461 nm) there is a dip in the amount of light that passes through the cell, meaning the Sr is absorbing the light as we expect. If the experimental and theoretical models don't match up, we will have to increase or decrease the laser frequency accordingly. Additionally, we will want to conduct this test at multiple different temperatures (400-500 ^{o}C) because at higher temperatures there will be more Sr in gaseous phase that should absorb more light. If not as much light is being absorbed as we want, we will test up to 540 ^{o}C. After ensuring we are getting a good absorption signal, we will want to conduct a long-term test to make sure Sr doesn't stick to the windows of the cell. Running SAS on the cell for at least a week straight, we should be able to visually see if any Sr begins to build up on the windows and run the laser frequency variation test explained above as well. During this time, we will also adjust the temperature of the sapphire windows to find a reasonable temperature that prevents Sr build-up.

Ultimately, I will then take all the techniques I learned, such as collecting data, analyzing absorption spectroscopy, and designing/constructing vapor cells, to work with a similar system using Rb instead of Sr in Professor Odom's lab. This system will similarly lock a laser to a frequency to properly excite Rb and measure its properties, but instead of using MOT, we will use an inert solid neon crystal to trap/isolate Rb. However, I am unlikely to make the transition to Odoms lab until the end of summer or fall quarter. If the transition ends up being set to occur during the summer term under WCAS Summer Grant, I will first seek approval by my faculty advisor (Odom) and the Office of Undergraduate Research.

At the start of the summer, I will have already completed a quarter of research with the Kovachy group working on this same project during the design phase. I have already been meeting weekly with Professor Kovachy and a grad student in his lab to better learn about the project and get guidance from them through this process. Additionally, I will have taken three quarters of Physic-126 (Physics lab for ISP), which has and will continue to prepare me with a general understanding of analyzing data. I have previously analyzed absorption spectroscopy in the form of infrared spectroscopy (similar to SAS but works more with molecules) in Chem-232-1 (Organic Chemistry lab for ISP and Chemistry Majors), which, while not the exact same, has given me some experience in analyzing absorption spectra. Additionally, to further my understanding of atomic physics, over spring quarter, besides gaining information directly from working in Kovachy's lab, I will read *Atomic Physics* by Cristopher Foot, suggested to me by both Professor Kovachy and Professor Odom. For the design and building phase of the project, if the need arises, I have completed twenty hours of instructional Milling and Lathing in the Research Shop, so if minor problems occur in the building phase, I should be able to help in modifying parts. Additionally, I plan to use all the techniques I'll learn over summer in Kovachy's lab to work on a similar project in Odom's lab, and this will hopefully provide a path to continue research throughout my undergraduate career in the field of physics.

Appendix I: References

[1] Godone, A., Levi, F., Calosso, C.E. *et al.* "High-performing vapor-cell frequency standards." *La Rivista del Nuovo Cimento.* 38, (2015): 133–171

[2] Neuman, J. A., P. Wang and Alan Gallagher. "Robust high-temperature sapphire cell for metal vapors." *Review of Scientific Instruments*. 66 (1995): 3021-23

[3]Stellmer, Simon, Florian Schreck and Thomas C. Killian. "Degenerate quantum gases of strontium." *arXiv: Quantum Gases.* (2011): 209-13

Appendix II: Covid back-up plan

A majority of my project involves data analysis, so if I happen to get covid, most of this work can be done asynchronously. Data measurements can be sent to me from the lab and I can join any group meetings or advisor meetings over Zoom. Additionally, since I am starting near the earlier end of the summer, if the need truly arises, I likely could continue this project and work with the lab for an additional week past the planned end date.